

Jet Kinematics of the Microquasar GRS 1915+105 during its Persistent X-Ray-Obscured State

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Abstract

Since 2019, the Galactic microquasar GRS 1915+105 has resided in a deeply X-ray-obscured state [1, 2]. To investigate how this dense, near-Eddington environment influences relativistic jet trajectories, we present a 12-epoch S-band MeerKAT radio monitoring campaign tracking symmetric bipolar jet lobes throughout 2025 [3]. Quantitative Bayesian model selection robustly favors a 1st-order ballistic model with a fixed position angle over precessing or accelerating alternatives. Assuming a distance of 9.4 ± 1.0 kpc [4], MCMC posterior sampling constrains the intrinsic jet velocity to $\beta = 0.41 \pm 0.08$, the bulk Lorentz factor to $\Gamma = 1.10 \pm 0.04$, and the jet inclination angle to $72.93^\circ \pm 4.50^\circ$. This velocity is fundamentally slower than historical unobscured jet speeds, providing direct kinematic evidence of a mass-loaded “slow-jet state” within the obscured accretion regime [5]. Furthermore, our derived jet inclination demonstrates a structural recovery from the extreme, nearly edge-on tilt of $87^\circ \pm 3^\circ$ observed in late 2023 [6].

Introduction

Historically, the Galactic microquasar GRS 1915+105 was characterized by highly relativistic, superluminal jet ejections ($\beta\Gamma \geq 1$) launched along a stable axis [7, 4]. This behavior changed dramatically in 2018 and 2019 when the system transitioned into an unprecedented, persistent X-ray-obscured state [1, 2]. Marked by heavily suppressed X-ray emission alongside elevated mid-infrared and radio flaring, this epoch is theorized to be driven by a severe accretion disk warp that has tilted Compton-thick material directly into the line of sight, shielding the central engine [8, 9].

Recent observations during this obscured phase point to a distinct shift in jet properties, characterized by lower intrinsic speeds ($\beta\Gamma \leq 0.40$) and significant variations in position angle [10, 11]. This supports an emerging paradigm: fast, highly relativistic jets are launched in low-density, unobscured environments, whereas heavily obscured environments—potentially driven by super-Eddington accretion—physically mass-load the outflow close to its launch site [5]. This acts as a drag mechanism that slows the resulting ejections to mildly relativistic velocities [5]. To investigate this slow-jet paradigm and map the geometry of GRS 1915+105 during this obscured phase, this study aims to:

- **Identify and track** individual jet components across consecutive observational epochs to capture their spatial evolution over time.
- **Determine** the precise proper motions of these components through quantitative kinematic modeling to extract the intrinsic velocity, constrain the jet inclination angle, and evaluate the physical coupling between the accretion environment and jet production.

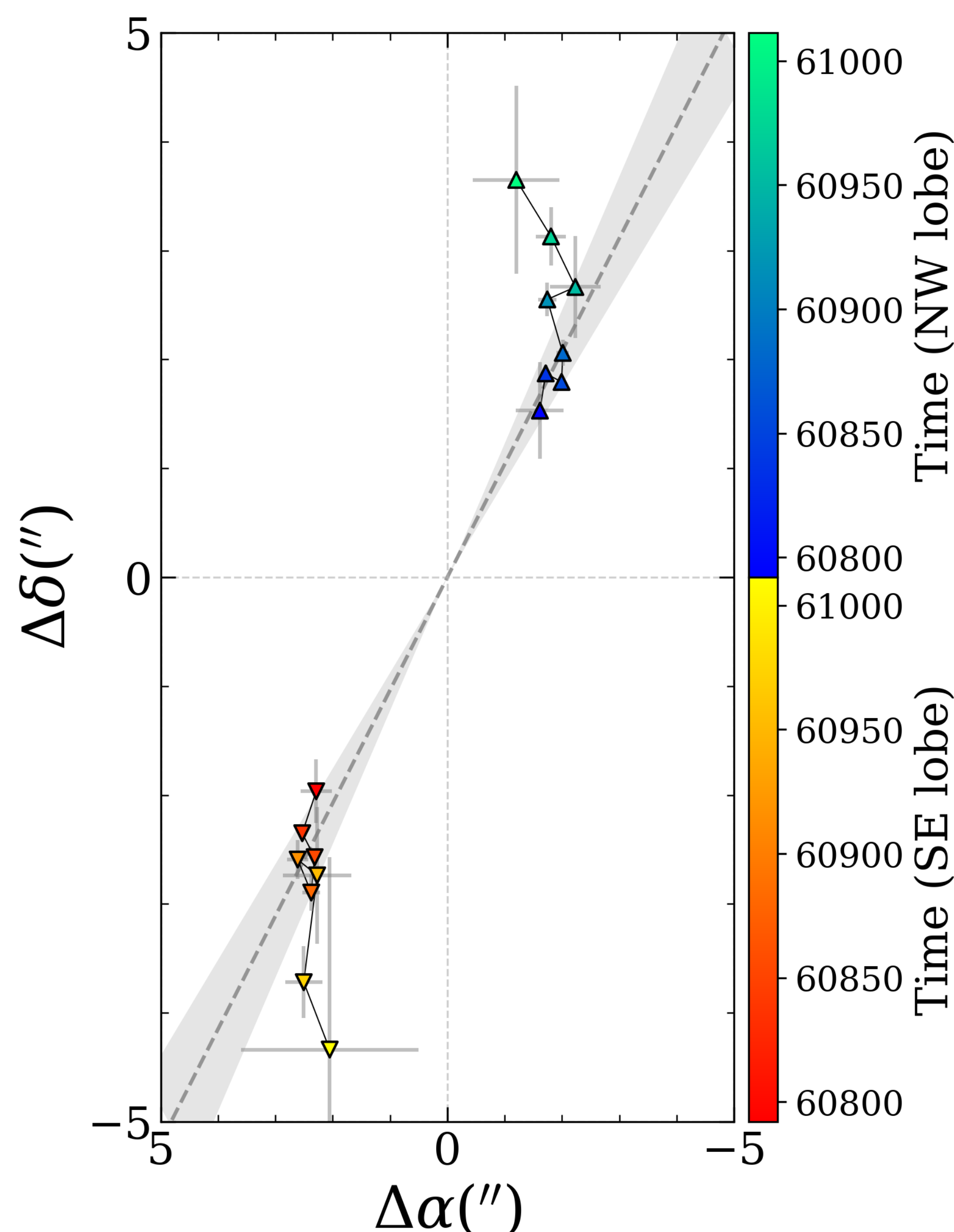


Figure 1: This is the caption describing your plot or figure.

Dataset and Kinematic Modeling

To track the spatial evolution of the jet ejecta, we utilized observations from the JetKAT project, comprising 12 primary S-band (2.3–2.8 GHz) MeerKAT observation blocks executed between January and December 2025 [3]. These were supplemented by five additional historical and concurrent blocks to provide an extended temporal baseline for the kinematic analysis [3].

To derive the intrinsic relativistic kinematics of the resolved bipolar lobes, we evaluated a suite of geometric models tracking the angular separation of the components over time. We tested constant velocity (1st-order ballistic) trajectories against accelerating (2nd-order polynomial) motions, paired with either fixed or precessing position angles. Quantitative model selection via the Bayesian Information Criterion (BIC) definitively favoured the 1st-order ballistic model with a fixed position angle over the more complex accelerating and precessing alternatives.

Employing the statistically preferred model, we performed Markov Chain Monte Carlo (MCMC) sampling to extract posterior distributions for the ejection parameters. The kinematic fits determined an extrapolated launch time of MJD 60438.42. Assuming a source distance of 9.4 ± 1.0 kpc [4], the apparent proper motions of the approaching (Southern) and receding (Northern) lobes translate to an intrinsic jet speed of $\beta = 0.4095^{+0.0762}_{-0.0849}$ and a bulk Lorentz factor of $\Gamma = 1.0961^{+0.0344}_{-0.0475}$. The jet incli-

nation angle is constrained to $\theta = 72.93^{+4.97}_{-4.01}$ to the line of sight. Furthermore, the position angles of the Northern and Southern lobes remained stable at -40.22° and -42.73° , respectively.

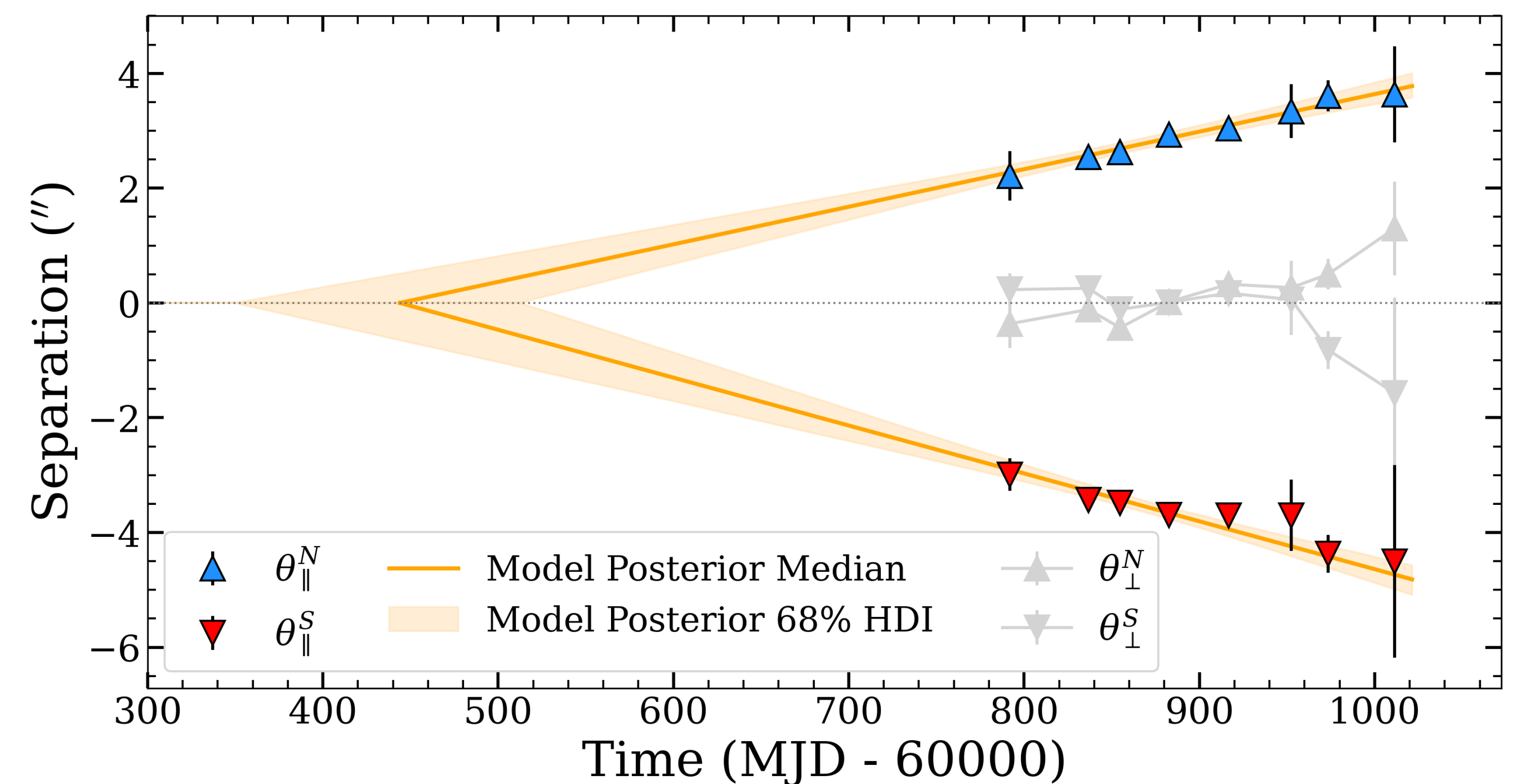


Figure 2: This is the caption describing your plot or figure.

Table 1: Kinematic Properties of the GRS 1915+105 Bipolar Jets

Parameter	Symbol / Units	Value (68% Credible Interval)
<i>Global Ejection & System Parameters</i>		
Extrapolated Launch Time	T_{ej} (MJD)	$60438.42^{+94.85}_{-65.59}$
Assumed Distance	D (kpc)	9.4 ± 1.0 [4]
Intrinsic Jet Speed	$\beta = v/c$	$0.4095^{+0.0762}_{-0.0849}$
Lorentz Factor	Γ	$1.0961^{+0.0344}_{-0.0475}$
Jet Inclination Angle	θ ($^\circ$)	$72.93^{+4.97}_{-4.01}$
<i>Receding Lobe (NW Component)</i>		
Proper Motion Speed	μ_r (arcsec/day)	$(6.4604 \pm 1.21) \times 10^{-3}$
Position Angle	PA_n ($^\circ$)	$-40.22^{+1.94}_{-2.17}$
<i>Approaching Lobe (SE Component)</i>		
Proper Motion Speed	μ_a (arcsec/day)	$(8.2326 \pm 1.55) \times 10^{-3}$
Position Angle	PA_s ($^\circ$)	$-42.73^{+1.67}_{-1.55}$

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